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Faculty Search Committee, Physical AI
Bradley Department of Electrical and Computer Engineering
Virginia Tech
Blacksburg, VA 24061, United States

Dear Members of the Faculty Search Committee,

Application for the Tenure-Track Faculty Position in Physical AI, Bradley Department of Electrical and Computer Engineering

I am writing to apply for the tenure-track faculty position in **Physical AI** in the Bradley Department of Electrical and Computer Engineering. I am currently a Research Assistant at the **Power Electronics and Machines Centre (PEMC), University of Nottingham**, having completed my **PhD in Electrical Engineering at Southeast University (SEU), Nanjing**, in September 2025 with the dissertation *Multi-physics Analysis and Design of Hairpin Winding PM Traction Machines in Electric Vehicles*. My research develops **physics-grounded, hybrid AI methods for modeling, simulating, and optimizing electrified power systems**, which maps directly onto the department's vision for AI grounded in first principles and experimental reality.

The core of my program is **physics-informed surrogate modeling**. I developed an **implicit-physics deep-learning surrogate** that maps device design parameters to the full magnetic vector potential field, reaching R^2 = 0.9996 for average torque and 0.9730 for iron loss while reducing inference time by more than **94%** relative to direct finite-element analysis. This is a concrete instance of **hybrid modeling that combines governing equations, simulation, and data**, and it is built on an **in-house Python finite-element code package** I wrote for coupled electromagnetic, structural-modal, and mechanical-stress analysis. The same framework supports rapid design-space exploration and **AI-assisted multi-objective optimization**, and it extends naturally to operator-learning and neural-surrogate methods across other governing equations relevant to ECE.

A second strand of my work bridges **theory, computation, and real-world data**. I developed high-fidelity lumped-parameter thermal networks and fluid-solid coupled cooling models for power-dense machines, **validated against AVL dynamometer measurements**, giving the hardware-anchored reduced-order models that underpin **digital-twin and condition-monitoring** workflows. Through a multi-year collaboration with **Inovance**, a world-leading EV traction-machine supplier, I delivered two oil-cooled traction machines (18,000 rpm, 113 kW; 20,000 rpm, 270 kW) through the full design-build-test cycle, from multi-physics design and engineering-drawing release to dynamometer characterization. This experience in **power electronics and power systems**, one of the department's named domain applications, is what I would extend toward **autonomous experimentation and self-driving laboratories** for electrified hardware.

Looking ahead, I would build a research program on **physics-informed and operator-learning surrogates**, **digital twins that fuse first-principles models with experimental data**, and **AI-accelerated Design-Build-Test-Learn** for power-electronic and electromagnetic systems. My combined academic and industrial record positions me to pursue competitive extramural funding from **NSF**, **DOE**, and **ARPA-E**, and to collaborate across Virginia Tech, including the Institute of Advanced Computing in Alexandria.

On teaching, I would be glad to contribute to **core computer-engineering courses such as embedded systems and computer architecture**, drawing on my experience writing **C/C++ embedded firmware for TI C2000 and STM32 controllers** and prototyping power converters, as well as specialized graduate courses in **physics-informed machine learning, multi-physics modeling, and electric drives**. I am committed to inclusive and supportive mentoring, and to fostering an educational environment consistent with the Virginia Tech Principles of Community.

My CV, three representative publications, research and teaching statements, statement on supporting an inclusive educational environment, and list of references are attached. I would be glad to discuss my application further.

Yours sincerely,

Wentao Zhang